

Customer Segmentation & Churn Pattern Analytics in European Banking

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1. Abstract

While many banks track generalized churn metrics, aggregate data fails to identify actionable interventions. This exploratory data analysis (EDA) applies systematic customer segmentation to a dataset of 10,000 European bank customers. By evaluating churn across geographic regions, age brackets, and financial tiers, this paper identifies high-risk clusters and quantifies revenue vulnerabilities.

2. Data Ingestion & Segmentation Methodology

Non-analytical fields (e.g., Surname, CustomerId) were removed to streamline analysis. To facilitate granular profiling, continuous variables were binned into discrete strategic segments:

- **Age Segmentation:** Sub-divided into <30, 30–45, 46–60, and 60+ to isolate generational banking behaviors.
- **Financial Profile:** Accounts were segmented into Zero-balance, Low-balance (<€50k), and High-balance (>€50k).
- **Tenure & Credit:** Segmented to evaluate long-term loyalty and creditworthiness against exit rates.

3. Churn Distribution & Geographic Risk

Overall baseline churn sits at approximately 20.37%. However, the Geographic Risk Index highlights severe regional disparities. Customers in Germany churn at roughly double the rate of those in France and Spain. Furthermore, an interaction analysis of Geography and Age reveals that German customers in the 46-60 age bracket represent the most acute demographic risk cluster in the entire dataset.

4. High-Value Customer Analytics

A scatter plot analysis of Estimated Salary versus Balance revealed that churn is not strictly a low-income or zero-balance phenomenon. In fact, a significant density of churners exists in the High-balance segment (accounts holding over €50,000). The loss of these premium customers disproportionately damages institutional liquidity, signifying that current retention efforts are failing to satisfy affluent clients.

5. Conclusion and Strategic Recommendations

By moving beyond aggregate churn rates and deploying segmentation analytics, we have identified critical vulnerabilities in the bank's customer lifecycle. To stabilize revenue, the institution must deploy immediate interventions for the German market, audit product offerings for senior citizens (46+), and assign dedicated relationship managers to high-balance accounts to prevent further capital flight.